

Phase 3: Project Design Phase

Project title : Script-Controlled ACL - Restrict Record Access Based on Field Value

College Name: Government Arts and Science College, Manur.

College Code: MSU142

Department: B.SC Computer Science

Team ID: SWTID-2026-9029

TeamLeader: Subbulakshmi M NM ID:727F7AC2ABD1D8C8DF6496517D478773

Team member: Arumuga Kani S NM id: 601EE17CA3AAD3AB4EA4C8F841A16DD1

Team member: Mariammal C NM id: BBE8324EB82C757A1054DE3C9951281A

Team member: Esakki Ganesh P NM id: A705CB2C71F3AF38AF2473EFB5AE7725

Team member: Akalya R NM id: B3F460E08A938D39C7F40868EA8EC1C6

1. Introduction

Project Design defines the structure and technical approach for implementing the Script-Controlled ACL system. The system is designed to restrict access to records based on the value of a specific field.

The design describes the system architecture, data structure, access-control logic, process flow, user interface, security mechanisms, and testing approach.

2. Design Objectives

The main objectives are:

- Design a secure record-level access-control mechanism.
- Restrict unauthorized users from accessing protected records.
- Use a field value as a condition for access.
- Implement the access rule using a server-side script.
- Provide clear authorized and unauthorized access behavior.
- Make the solution easy to test, maintain, and demonstrate

3. High-Level Architecture

The proposed system consists of the following components:

1. User

- Attempts to access a record.

2. Application / Platform

- Receives the user's request.
- Identifies the requested record.

3. Access Control Layer

- Checks the applicable ACL rule.

4. Script Logic

- Reads the required field value.
- Compares it with the access condition.
- Returns an authorization result.

5. Database / Record

- Provides the requested record only when access is permitted.

Architecture Flow

User → Application → ACL → Script Condition → Access Decision → Record

If the condition is satisfied, access is granted. Otherwise, access is denied.

4. Component Design

4.1 User Component

The user interacts with the application and attempts to view or access records.

4.2 ACL Component

The ACL controls whether the user is authorized to access the requested record.

4.3 Script Component

The server-side script evaluates the field value of the record and determines whether the current user should be allowed to access it.

4.4 Data Component

The data component contains the records and the field whose value is used for access control.

4.5 Testing Component

Test users and records are created to verify both successful and unsuccessful access attempts.

5. Data Design

A sample record structure can contain:

Field	Description
Record ID	Unique identifier of the record
Record Name	Name or description of the record
Access Field	Field used for access validation
Owner/User	User associated with the record
Status	Current record status
Created Date	Record creation date

The Access Field is the key field used by the ACL script.

6. Access-Control Design

The system follows a conditional access model.

Access Logic

- User requests a record.
- ACL is triggered.
- Script reads the required field.

- Script compares the field value with the configured condition.
- If the condition is satisfied, access is allowed.
- If the condition is not satisfied, access is denied.

Conceptual Logic

IF record field value satisfies access condition

THEN allow access

ELSE

deny access

The exact field and condition can be configured according to the project requirement.

7. Access Matrix

User/Condition	Matching Field Value	Non-Matching Field Value
Authorized User	Access Allowed	Access Denied
Unauthorized User	Access Denied	Access Denied
Administrator	Based on configured permissions	Based on configured permissions

The final behavior depends on the ACL configuration and platform security rules.

8. Process Design

Record Access Process

Step 1: User logs into the application.

Step 2: User attempts to open a record.

Step 3: The ACL associated with the record is evaluated.

Step 4: The server-side script retrieves the required field value.

Step 5: The script evaluates the access condition.

Step 6: The system makes an authorization decision.

Step 7: The record is displayed if access is permitted; otherwise, access is blocked.

9. User Interface Design

The user interface should provide:

- Login/authentication screen.
- Record list or record access screen.
- Record details page.
- Appropriate access-denied message.
- Clear indication of successful access.

The interface should remain simple so that the access-control functionality can be demonstrated clearly.

10. Security Design

Security is the main focus of the project.

The design includes:

- Server-side access validation.
- ACL-based authorization.
- Field-value validation.
- Prevention of unauthorized record access.
- Separate testing of authorized and unauthorized users.
- Avoiding reliance only on client-side restrictions.

The authorization decision should be performed on the server side so that users cannot bypass the restriction by modifying client-side information.

11. Error Handling

The system should handle invalid or unexpected situations appropriately.

Examples include:

- Missing field value.

- Invalid record.
- Unauthorized user.
- Incorrect access condition.
- Script execution error.

For unauthorized access, the system should prevent the record from being displayed and provide an appropriate access-denied response.

12. Test Design

The design supports the following test cases:

Test Case 1 – Authorized Access

A user satisfying the configured access condition attempts to access a record.

Expected Result: Access is granted.

Test Case 2 – Unauthorized Access

A user who does not satisfy the access condition attempts to access a record.

Expected Result: Access is denied.

Test Case 3 – Different Field Value

The record contains a different value in the controlled field.

Expected Result: The ACL evaluates the new value and applies the correct access decision.

Test Case 4 – Missing Field Value

The controlled field is empty or unavailable.

Expected Result: Access is handled securely according to the configured rule.

13. Phase 3 Deliverables

The Project Design phase produces:

- System architecture.
- Component design.
- Data design.
- ACL design.
- Access-control logic.

- Process flow.
- UI design.
- Security design.
- Error-handling approach.
- Test design.

14. Conclusion

The Project Design establishes the technical structure for the Script-Controlled ACL – Restrict Record Access Based on Field Value project.

The design uses an ACL and server-side script to evaluate a field value before allowing access to a record. This provides a structured approach to implementing, testing, documenting, and demonstrating field-based record access control.